Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 02
Prof. Dr. Simon Nestler

Building a gesture set and breaking it

What is this sheet about?

This sheet moves from diagnosing other people's gesture sets to building your own. Your team trains a recognizer on gestures you define, then lets people from outside the team use it – and watches where it fails. Recognition rates are data, not a verdict: what counts is why a gesture was misunderstood and what that costs the person using it.

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Docking point: This exercise cashes in sessions T04 (*The gesture the system misunderstands*) and T05 (*Hands in the air*): in T04 you diagnosed why other people's gesture sets fail — today you build one yourself and let strangers show you why yours does.

Organisation: Please work in your project team of 3–4 students (the team fixed in exercise 1). Pick **one** recognizer platform per team: *Teachable Machine* (browser, webcam: train an image or pose classifier on your own gestures in minutes; the fast, rough option) or the *MediaPipe Gesture Recognizer* (Google AI Edge, web: the built-in gesture vocabulary plus hand-landmark output; the more controllable option). Both are free and run in the browser; neither requires code, though MediaPipe rewards a little JavaScript. You also need one phone or laptop per team for filming test sessions, **with the tested person's consent; ask first, every time**.

AI role — tool (declared) for training the recognizer and analyzing your data. The guessability test itself runs without AI: humans do the guessing, unprompted. Your upload must disclose what was made with which tool.

Bonus points: In order for the bonus rule to apply (participation in all six blocks with an upload within 48 hours each earns 5 % compensation on the exam), your team must hand in one complete submission package (see the end of this sheet) within 48 hours after this exercise. This is block two of six.

First assignment (20 minutes): Define the vocabulary (team). Choose 4–6 commands (referents) from your project idea and design one gesture for each. Write down, per gesture: registration, continuation, termination (the grammar from T04). Then answer the T04 stress question, in writing: **which everyday movement could fire this gesture when nobody means it?**

Second assignment (40 minutes): Train the recognizer (team, split roles). Two people train with multiple performers, varied backgrounds and lighting: **a recognizer trained on one person in one corner recognizes exactly that**. One or two team members prepare the third assignment in parallel. Target: at least three gestures recognized live, and one honest sentence about the ones that are not.

Third assignment (15 minutes): Design the guessability protocol (team). For each command, prepare a referent card that shows only the desired *effect* (text or sketch, **never the gesture**). Define what you record per trial: guessed gesture, match yes/no, hesitation, comments. This is the T04 miniature scaled up to a real protocol.

Fourth assignment (40 minutes): Field test with strangers (pairs, outside the room; leave the building if you must). Test **at least three people who are not part of this course**: the cafeteria line, the library, the hallway. Show each person the referent cards one by one and ask them to *perform* the action they would expect; film with consent or note by hand. One tester runs the session, one records. **Do not teach, do not correct, do not helpfully wiggle your own hand.**

Fifth assignment (25 minutes): Analyze and revise (team). Compute the plain agreement per referent: how many of your testers produced your gesture? Identify the worst performer and revise it — new gesture, or a documented reason to keep it despite the data (there are legitimate ones; "we liked ours" is not among them). If time allows, retrain the recognizer on the revision.

Sixth assignment (10 minutes): Upload preparation. Package the submission while the data is fresh.

Buffer and critique (30 minutes): Teams return at minute 150 sharp. Quick round: each team names its most-guessed and least-guessed referent and one sentence on what surprised them. The buffer also absorbs slow recruiting. Finding three strangers takes longer than you think.

The submission (within 48 hours, one upload per team, via Moodle) consists of:

- `gesture-set.md` — the 4–6 gestures with registration/continuation/termination each, plus the false-positive risk sentence per gesture (max 2 pages).
- `guessability-protocol.md` — method, number and rough description of testers (no names needed), agreement per referent, the revised gesture with reasoning (max 3 pages).
- `recognizer/` — Teachable Machine share link or MediaPipe configuration, plus a screen recording of live recognition (max 90 seconds, .mp4, max 100 MB).
- Optional: consented test clips (max 3 × 30 seconds).
- **AI disclosure line** (in `gesture-set.md`, mandatory): *"Made with AI: [tool] was used for [part]; checked and understood by [name]."* No AI used is also a disclosure.

Put everything into one upload per team and hand it in via Moodle within 48 hours after this exercise.